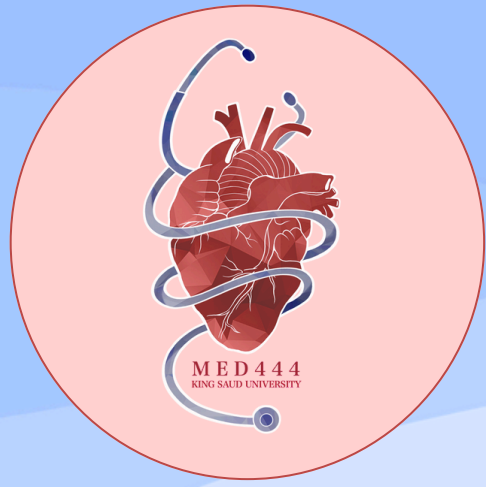
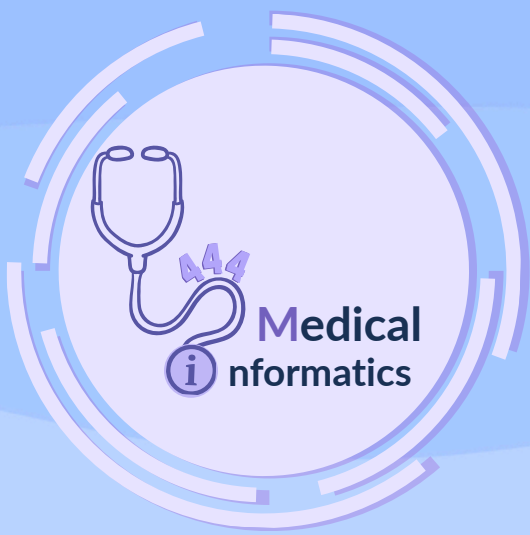




Electronic health records and interoperability

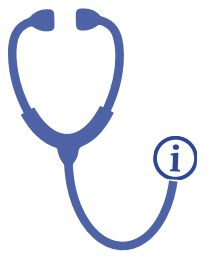
Editing files

- main text- male slides
- female slides- important
- dr notes - textbook
- extra / old slides

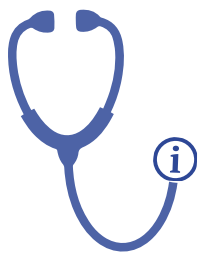
objective



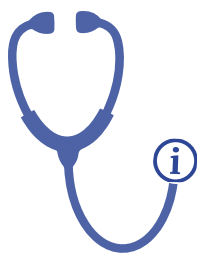
* State the definition of electronic health records (EHRs)



* Describe the limitations of paper-based health records



* Identify the benefits of EHR



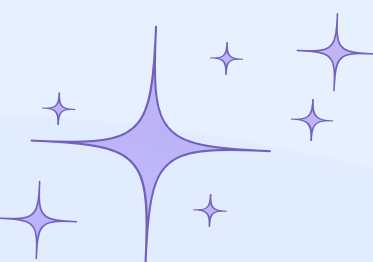
* List the key components of an EHR



* Describe the benefits and challenges of computerized order entry and clinical decision support systems



* Enumerate the steps to adopt and implement an EHR



اللهم انفعني بما علّمتني، وعلمني ما ينفعني، وزدني علماً

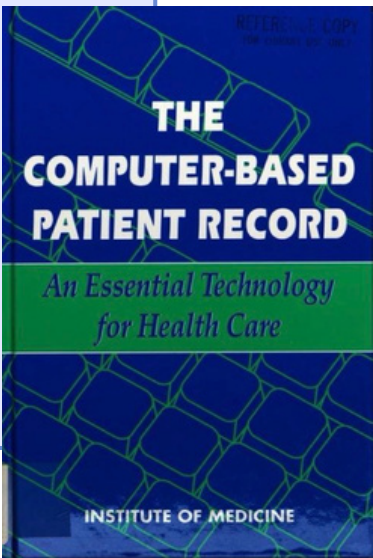


Introduction

There is no topic in health informatics as important, yet controversial, as the electronic health record (EHR)

In spite of fledgling EHRs being around for the past 35-40 years they are still controversial in the eyes of many

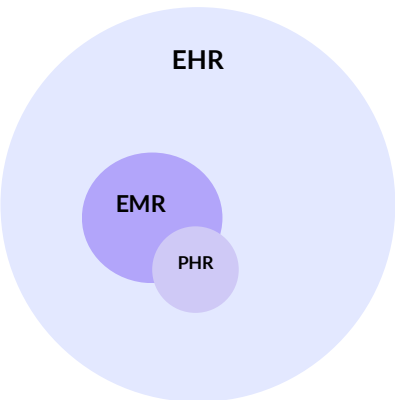
In 1991, IOM (**Institute of medicine**)recommended EHR as a solutionf or many problems.
The Computer-Based Patient Record: An Essential Technology for Health Care.



- In the book they outlined eight core functions all EHRS must have:
1. Health information and data
 2. Results management: Physicians should not have to search for lab, x-ray and consult results -can be accessed easily-
 3. Order management (CPOE)
 4. Decision support : Providing alerts and reminders. E.g. reminds clinicians about drug-drug interactions and known medication allergies
 5. Electronic communication and connectivity: Communication among disparate partners include all tools such as secure messaging, text messaging, web portals, health information exchange, etc.
 6. Patient support: Recognizes the growing role of the Internet for patient education as well as home telemonitoring.
 - 7.Administrative processes and reporting: Electronic scheduling, electronic claims submission, eligibility verification, automated drug recall messages, automated identification of patients for research.
 8. Reporting and population health: Reporting of immunization status and biosurveillance data to an electronic format.
- Experts suggest several future trends of EHR, including an increased reliance on cloud computing.

Definitions

Electronic Health Record There is no a universal accepted definition of EHR	"An electronic record of health-related information on an individual (They can be healthy not necessarily patients) that conforms to nationally recognized interoperability standards and that can be created, managed and consulted by authorized clinicians and staff across more than one healthcare organization "
Electronic Medical Record	"An electronic record of health-related information on an individual that can be created, gathered, managed and consulted by authorized clinicians and staff within one healthcare organization. " The difference between EMR & EHR is that EMR is for one health organization but EHR for more than 1 organization
Personal Health Record	"An electronic record of health-related information on an individual that conforms to nationally recognized interoperability standards and that ca be drawn from multiple sources while being managed, shared and controlle by the individual. " (The patient here is responsible for this record unlike EHR and EMR where th staff and the organization are responsible)



The relationship between EHR, PHR, and EMR.

In some textbooks, the terms “EHR” and “EMR” are used interchangeably, even though there’s a great difference between them.
EHR will handle information of both patients and healthy people. EHR is NOT a digital copy of a paper record. PHR will be controlled by the individual instead of hospitals or healthcare organizations.

IMPORTANT!
Understand the difference between the terms and major keywords

Why do we need EHRs?



1

Paper records are severely limited

Paper records: illegible handwriting, limits the ability to extract and use information, I can't share it or store it, expensive to copy and transport, easy to destroy, Difficulty traced -difficult to determine who has seen it- EHR: easily coded, available off-site 24\7, and identify eligible patient for research.

2

Need for improved efficiency and productivity

EHRs allow for the reduction in duplication of tests ordered for pts (because the data is there , so we can check

whether the test was done for the patient or not), prescriptions, allow remote scheduling and payment, EHRs use

templates, click down, drop-down menus which reduces redundant administrative paperwork.

3

Quality of care and patient safety

Improves eligibility of clinical notes, improve access, reminders of preventive services, availability of summary list.

Eases the collection of quality reports. Improve patient safety through multiple mechanisms: (1) Improved legibility of clinical notes.(2) Improved access anytime and anywhere.(3) Reduced duplication. (4)Reminders and clinical alerts (for example a

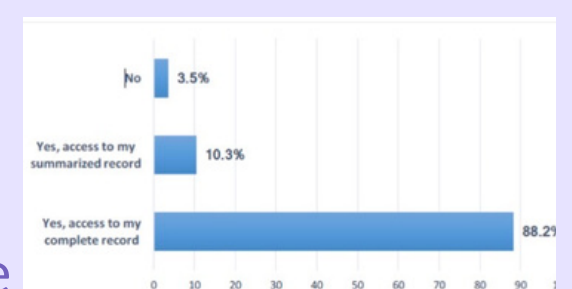
reminder that announces if relevant tests or preventive services are overdue). (5) Clinical decision support that reminds clinicians about drug-drug interactions, known medication allergies, cost and correct dosage of drugs. (6)Electronic problem

summary lists (PSLs) provide diagnoses, allergies and surgeries at a glance.

4

Public expectations

General public favors EHR as it increases their overall satisfaction, fewer lost charts, faster refills and improved delivery of patient educational material, allow patients access to their records with multiple other functionalities such as online appointments.



5

Financial savings

As by paper needs storage rooms, printing etc it has a financial cost, this point is considered a both a pro and a con as medical and legal financial losses can happen during the transition between paper based systems and EHRs "implementation chasm". Reduction in the need for transcriptionists with the advent of point and-click templating and voice recognition software, fewer callbacks from pharmacists requiring help to decipher physician handwriting. The labor costs of chart pulls are reduced.

6

Technological advances

7

Need for aggregated data

As we have a lot of external and internal data, aggregation Can't be done with paper, paper can't be integrated, we also need aggregated data because we can analyse it.

Digital healthcare information can be integrated with multiple internal and external applications: 1. Integrate with health information organizations (HIOs). 2. Integrate with analytical software for data mining to examine optimal treatments.

3. Integrate with local, state and federal government. 4. Integrate with artificial intelligence.

5. Integrate with personal devices, such as activity monitors, glucose monitors, etc.





8

EHR as a transformational tool

Allows analysis therefore improves healthcare. And Because it uses standardized health care (guidelines).

The integration of data analytics with care has resulted in the improvement in standardization of care, care coordination of groups within a hospital or from different hospitals.

9

Need for coordinated care

Either within the institution itself between its teams or between multiple institutions as an example; Home

monitoring (telehomecare) can transmit patient data from home to an office's EHR so the physician can check and make future suggestions.

EHRs key components

• Clinical decision support

Guidelines, reminders and alerts

• Secure messaging

For communication between patients and office staff and among office staff.

• Computerized physician order entry

• Practice management

software. scheduling software and patient portals that are embedded or connect with an interface of EHR.

• Referral management

• Results retrieval

• Prior encounter retrieval

• Patient reminders

• Ability to scan in data

An example of software that convert different type of documents into searchable and editable documents is OCR (optical character recognition).

• Ability to graph and track results

Ability to create graphs or flow sheets of lab results or vital signs.

• Ability to create patient lists

• Ability to create registries

• Electronic encounter notes

• Multiple input methods

(typing, dictation, voice recognition and templates).

• Access via mobile technology

• Remote access from home

• Electronic prescribing

• Integration with images

(PACS), a picture archiving and communication system (PACS) is a system that is used to capture those images.

• Integration with physician and patient education

• Public health reporting

• Problem summary lists

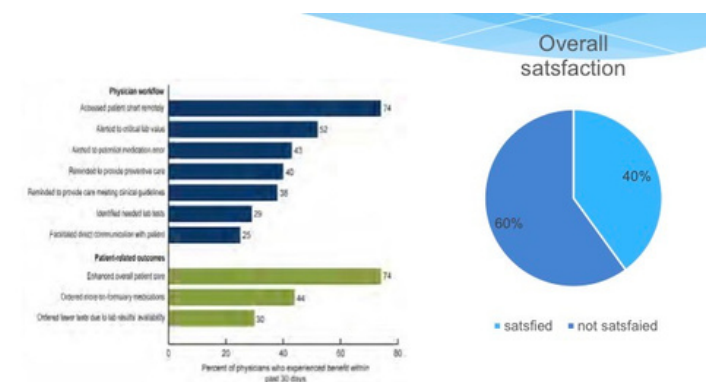
that includes: diagnoses, allergies, surgeries and medications.

• Privacy/security compliance

with HIPAA standards.

• Robust backup systems

• Support for client server or application service provider (ASP) modes

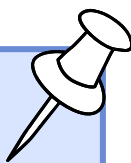


Computerized Physician Order Entry (CPOE)



CPOE

CPOE is an EHR feature that processes orders for medications, lab tests, imaging, consults and other diagnostic tests.



CPOE has many potential benefits:

- 1

Reduce Medication Errors

-Although it reduces medication errors, the degree of benefit is not that optimistic. This is due to new errors that result from technology “ e-iatrogenesis” these errors are more common in **outpatient** CPOE, because they are more in number.

-Examples of medication errors that were overcome by the use of CPOE:
the issue of illegibility that is associated with errors in ordering drugs with similar names, trailing zeroes,reduce under and over-prescribing,
- 2

Reduce costs
- 3

Reduce Variation of Care

(438) Specific guidelines that you follow to deliver standardized care.
- 4

Unintended adverse consequences

(= e-iatrogenesis)

Clinical Decision Support Systems (CDSSs)

“any electronic or non- electronic system designed to **aid directly in clinical decision making**, in which characteristics of individual patients are used to generate patient-specific assessments or recommendations that are then **presented to clinicians for consideration**.

Types of CDSs

1.Knowledge support	-Examples include: UpToDate, diagnostic (ICD-10) codes, and infobuttons (digital knowledge-based resources are being integrated with EHRs. For example, UpToDate has been embedded into several EHRs, and diagnostic (ICD-10) codes can be hyperlinked to further information or you can use infobuttons.Infobuttons are a HL7 standard and commonly used to link to important information).
2.Calculators	-appropriate antibiotic dosing
3.Flow charts and graphs	-to look at lab or vital sign trends over time (Trending/Patient tracking)
4.Medication order support	-(438)To detect any allergies or drug-drug interaction and also proper dosing.
5.Reminders	-remind clinician or patient about pending tests, etc. (yearly preventive health screening measures, Example: Mammogram)
6.Order sets	-inpatient clinical practice guidelines for specific scenarios. -Order sets are groups of pre-established related orders that are related to a symptom or diagnosis. For instance, with just a few mouse clicks a provider may place an order set for pneumonia that might include the antibiotic of choice, supplemental oxygen, an order for a chest x-ray.
7.Differential diagnosis	-software exists that helps clinicians analyze symptoms and signs, to arrive at a diagnosis
8.Lab and Imaging decision support	-particularly those in training, physicians may order imaging studies that are either incorrect or unnecessary. That is why Appropriateness criteria was integrated into the EHR which lead to reduction in ordered lab tests.
9.Public health alerts	-primarily infectious disease alerts for new outbreaks, e.g. MERS virus -The EHR-based alert also hyperlinked to disease specific order sets for educational tips, lab and medication orders.

EHR Registries

For example, you will collect data about diabetes so you can find the best way to treat your patients (clinical) but if you used this data to know how many patients are there so you can increase your staff to provide equal care for all then this (policy purpose).



An organized system that uses observational study methods to collect uniform data (clinical and other) to evaluate specified outcomes for a population defined by a particular disease, condition, or exposure, and that serves one or more predetermined scientific, clinical, or policy purposes.

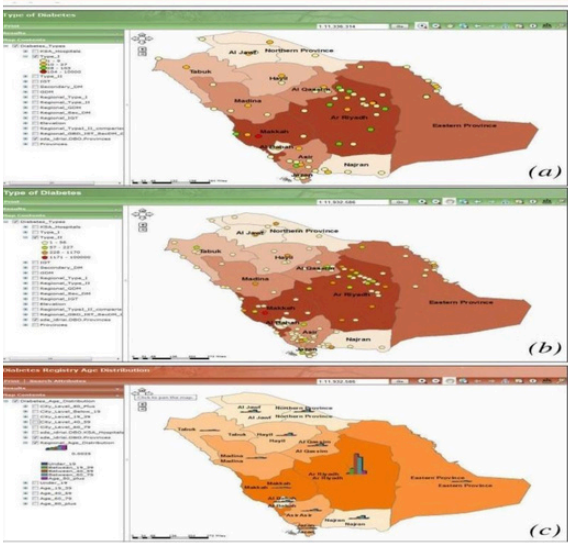
Types of EHR registries:



- Functions of EHR registries:
- 1- determining the natural history of diseases.
 - 2- ensuring effectiveness,safety and quality.

Example of the use of EHR registries: here they utilized the information from diabetes registry and GIS in decision making. Geographic information system (GIS) maps demonstrating the diabetic patient distribution for:

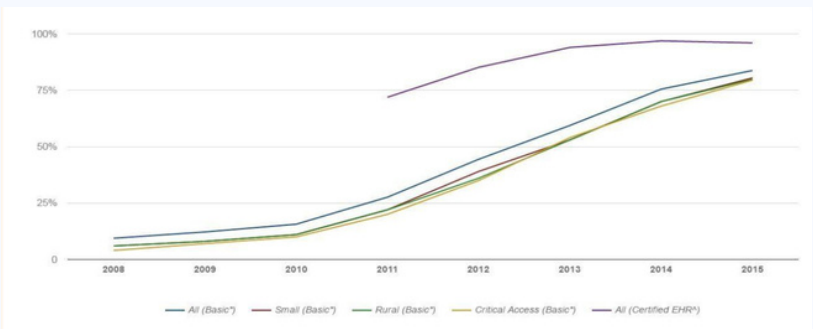
- o (a) type 1 diabetes
- o (b) type 2 diabetes at the country level
- o (c) the distribution of different age groups in all health sectors.



EHR Adoption

Inpatients EHR Adoption

Adoption of a certified inpatient EHR by non-federal acute care hospitals had risen from 72% in 2009 to 96% in 2015.



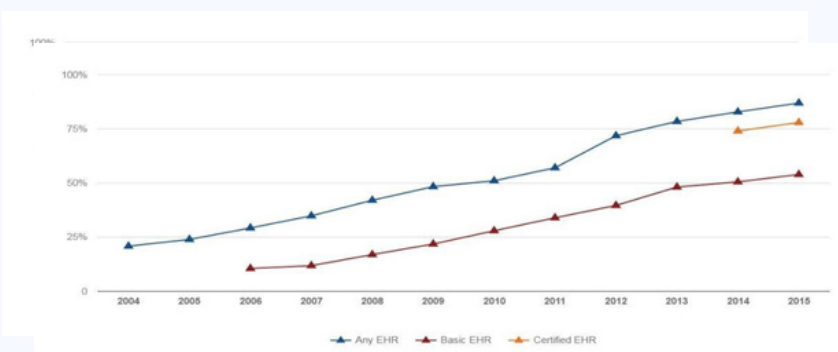
EHR adoption in Saudi Arabia
-Eastern Province study

(Bah, Alharthi, El Mhalli, 2011):

- o Only 3 of 19 hospitals adopted EHR. They implemented the same EHR system which includes three main modules; laboratory, radiology, and pharmacy.

Outpatients EHR Adoption

86% of OP (office-based care repondants) had an EHR ; 54% with a basic system (has core function) and 78% with a certified system (a specific organization in the us gives certification if EHR meets a certain criteria).



Riyadh study (aldosari, 2014).

- 85%of Riyadh hospitals had EHR and out of these (50% had fully-functioning EHR,36% in progress (meaning they had EHR for some modules only).14-15% didn't have EHR.

EHRs Adoption cont.



Implementing an EHR steps:

1-Pre-implementation:

begins with deciding whether to purchase an EHR and ends with signing a contract with a vendor for a specific EHR.

-Decision of purchasing EHR

-Workflow mapping (Aka: workflow analysis): a detailed step-by-step description, typically utilizing a flowchart of how a process is accomplished.

For example, how are notes created or how are patient messages handled or how are prescription refills managed?

3-Post- implementation (maintenance):

remains in effect for the duration of EHR use.

2-Implementation:

-starts with the signing of the contract and ends with the go-live date. -Experts in IT implementations often categorize facets of implementation into People (team), Process (tactics), or Technology issues.

-Team:

-Key people issues include leadership, change management, goal establishment and expectation setting. An implementation will have three key types of leaders: a project manager, a senior administrative sponsor, and a clinical champion. clinical champion: a respected, supportive, influential clinicians (doesn't have to be a physician) that encourage other physicians to accept and utilize the system effectively.

Project manager: knowledgeable and experienced in managing a complex IT project.

a senior administrative sponsor: essential because an EHR implementation will affect nearly all aspects (departments) of a hospital.

-Tactics: It's the process, implementation. How will we use the EHR to redesign our workflows? What is our data entry strategy? Which data will we enter discretely, which will we scan and which (if any) will we leave out of the EHR? Who will do this data entry and when? Will we follow a big bang (all personnel/sites and EHR functions at once) or a phased implementation approach (certain user groups and/or certain sites/ departments and or certain EHR functions in sequential order)?

-Technology.

EHR Challenges

1-Financial Barriers

Most common one.

- particularly for smaller or solo practices.
- Multiple surveys report that lack of funding is the number one barrier to EHR adoption.
- web-based application is a less expensive choice.

2-Physician Resistance

Because workflow changes and indeed require clinicians to significantly alter the way they deliver patient care.

3-Loss of Productivity

- Because instead of meeting patients the physician spends her\his time training on how to use the EHR.
- .This is a period when physician champions can help maintain morale and momentum with a positive attitude

4-Workflow Changes

We have to use workflow analysis.

Workflow analysis should be done before the implementation of EHR to determine optimal alterations to work flows to ensure uninterrupted and efficient patient care after implementation.

5-Reduced Physician-Patient Interaction

- No eye contact, so you can use tablet instead of PC and work on PC placement.

6-Integration and interoperability issues

Data standards such as HL7. (interoperability)

- Perhaps the most interesting standard with interoperability ramifications is the HL7 standard (to solve CDSS interoperability issues) known as Fast Healthcare Interoperability Resources (FHIR).
- Integration with genomic databases.
- commercial EHR products require integration with existing software, other EHRs, registries, health information networks, and data warehouse



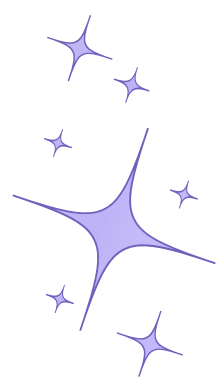
EHRs challenges cont.

7-Usability issues	● Usability has been defined as the “Effectiveness, efficiency and satisfaction with which specific users can achieve a specific set of tasks in a particular environment”. ● Effectiveness is the ability to do things, Efficiency doing things with minimal resource waste. ● Commercial EHRs might be different that healthcare environment. ● The American Medical Informatics Association (AMIA) board has made a set of recommendations to enhance patient safety and quality of care by improving the usability of EHRs: -Recommendations for the academic informatics community emphasizing human factors research. -Policy recommendations include standardizing EHR systems and ensuring interoperability, establishing -Specific recommendations for EHR vendors include developing a common user interface style. -Recommendations for end users suggests adopting best practices for EHR system implementation. (use the best EHR one that has worked successfully for other hospitals and is certified).
8-Privacy concerns	● Hacking into EHRs could result in loss of privacy for thousands, rather than a single paper chart. ● An important solution of security concern is using database or transmission encryption. ● Users of EHRs must: -Provide physical and software security of data systems - Provide physical and software security of their network(s) including mobile and remote computing. - Provide access control with defined user roles, - Monitor and manage user behavior. - Have written security policies and procedures. - Have an effective disaster recovery plan.
9-Legal	● It is not known if EHRs will increase or decrease malpractice over the long haul. ● E-iatrogenesis. Potential risks: -Clinical documentation. -Clinical decision support.
10-Inadequate proof of benefit	● In spite on many published studies, there is not adequate proof that EHRs improve quality of care.
11-Patient safety and unintended consequences	● Not only are studies suggesting improved patient safety mixed, there is evidence that new medical errors may occur (at least in the short term) with EHR use. ● “E-iatrogenesis”: medical errors due to technology (Over Dependent on technology can be challenging especially if we had system shutdown). ● Reliability issues: You should have a backup plan when the system is shut down; paper based records for example.

Conclusions:

- Paper based health records are severely limited.
- In spite of many potential benefits of EHRs, multiple challenges are associated with adoption.
- Planning, training and strategizing about EHRs is more important than the actual EHR brand purchased.

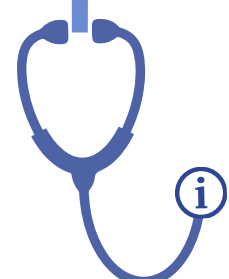




SUMMARY



Electronic Health Record (EHR)	An electronic record of health-related information on an individual that conforms to nationally recognized interoperability standards and that can be created, managed and consulted by authorized clinicians and staff across more than one healthcare organization.
Advantages	1-paper records are severely limited 2-need for aggregated data 3-quality of care and patient safety 4-public expectations 5-need for for improved efficiency and productivity 6-EHR as a transformational tool 7-need for coordinated care 8-technological advances
Key Components	1- Clinical decision support 2- Results retrieval 3- Access via mobile technology 4- Integration with physician and patient education 5- Secure messaging 6- Prior encounter retrieval 7- Remote access from home 8- Public health reporting 9- Computerized physician order entry (CPOE) 10- Patient reminders 11- Electronic prescribing 12- Problem summary lists 13- Practice management software 14- Electronic encounter notes 15- Integration with images 16- Ability to scan in data 17- Referral management 18- Multiple input methods 19- Robust backup system 20- Ability to graph and track results 21- Ability to create patient lists 22- Ability to create registries 23- Privacy/security compliance 24- Support for client server or application service provider (ASP) modes
Computerized Physician Order Entry	is an EHR feature that processes orders for medications, lab tests, imaging, consults and other diagnostic tests.
Clinical Decision Support Systems	Any electronic or non-electronic system designed to aid directly in clinical decision making. Types: Knowledge support, Calculators, Flow charts and graphs, Medication order support, Reminders, Order sets, Differential diagnosis, Lab and Imaging decision support, Public health alerts.
EHR Registries	-An organized system that uses observational study methods to collect uniform data (clinical and other) to evaluate specified outcomes for a population Types: Chronic disease management registries, Research registries, Safety registries, Public health registries, Quality registries
EHR Adoption	Steps: 1-Pre-implementation 2-Implementation 3-Post- implementation (maintenance)
EHR Challenges	Financial Barriers - Physician Resistance - Loss of Productivity - Workflow Changes - Reduced Physician - Patient Interaction - Usability issues - Integration and interoperability issues - Privacy concerns - Legal - Inadequate proof of benefit - Patient safety and unintended consequences





Q1 Which of the following best describes a potential unintended consequence of Computerized Physician Order Entry (CPOE)?

- | | | |
|---|--|--|
| A) Complete elimination of medication errors. | B) Improved legibility of orders, reducing errors. | C) Introduction of new technology-related errors (e-iatrogenesis). |
|---|--|--|

Q2: A hospital decides to adopt an EHR system and appoints a “clinical champion.” What is the primary role of this individual?

- | | | |
|--|--|---|
| A) Providing technical IT support during implementation. | B) Leading administrative budgeting decisions. | C) Influencing and encouraging other clinicians to accept and effectively use the EHR system. |
|--|--|---|

Q3: Which interoperability standard is specifically designed to improve data exchange and integration between Clinical Decision Support Systems (CDSS) and EHRs?

- | | | |
|-----------|-------------|---------|
| A) ICD-10 | B) HL7 FHIR | C) PACS |
|-----------|-------------|---------|

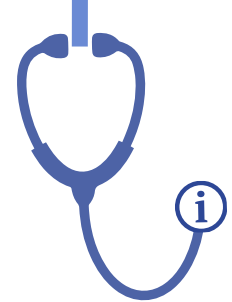
Q4: Why do we need export system in CDSS?

- | | | |
|-------------------------------|-------------------------|-------------------------|
| A- Preservation of knowledge. | B- Sharing of knowledge | C- Aid decision making. |
|-------------------------------|-------------------------|-------------------------|

Q5: Which one of the following is considered to be a legal EHR challenge?

- | | | |
|---|---|----------------------|
| A-Not known if EHRs will increase or decrease malpractice over the long haul. | B-Commercial EHRs might be different that healthcare environment. | C-Hacking into EHRs. |
|---|---|----------------------|

1-C 2-C 3-B 4-C 5-A





Team Leaders

Shahad Alshehri



Khalid Alghamdi

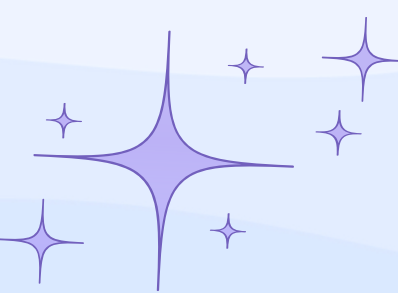


Team Members

Hessa Alamer



Nouf Alotaibi



سبحان الله، الحمد لله، لا إله إلا الله، الله أكبر،
لا حول ولا قوة إلا بالله

